Geometry, Triangles & Right-Triangle Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the legs are 5 and 12. What is the hypotenuse?

Answer: 13. Applying Area, Volume, Triangles, Circles & Trigonometry principles, $\sqrt{25+144} = \sqrt{169} = 13$ Therefore, the verified correct option is 13.

2 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 6 and 8. What is the length of the hypotenuse?

Answer: 10. By the Pythagorean theorem: $c = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

3 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a triangle with base 7 and height 10?

Answer: 35. Area $= \frac{1}{2}bh = \frac{1}{2}(7)(10) = 35$

4 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the side opposite angle θ has length 8, the side adjacent to θ has length 15, and the hypotenuse has length 17. What is $\cos(\theta)$?

Answer: 15/17. $\cos(\theta) = 15/17$ using SOH-CAH-TOA with sides $8, 15, 17$.

5 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 9 and 40. What is the length of the hypotenuse?

Answer: 41. $c = \sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = \sqrt{1681} = 41$

6 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a rectangle with length 6 and width 5?

Answer: 30. Applying Area, Volume, Triangles, Circles & Trigonometry principles, Area $= l \times w = 6 \times 5 = 30$ Therefore, the verified correct option is 30.

7 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A pole casts a shadow 3 feet long. The distance from the top of the pole to the tip of the shadow is 5 feet. How tall is the pole?

Answer: 4. By the Pythagorean theorem: $h = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$ feet.

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Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A circle has equation $(x-2)^2+(y+1)^2=16$. What is the radius?

Answer: 4. Applying Area, Volume, Triangles, Circles & Trigonometry principles, $r^2=16 \Rightarrow r=4$ Therefore, the verified correct option is 4.

9 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

Answer: 13. Applying Area, Volume, Triangles, Circles & Trigonometry principles, $\sqrt{25+144}=\sqrt{169}=13$. Therefore, the verified correct option is 13.

10 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the coordinate plane, point A is at $(1,2)$ and point B is at $(5,5)$. What is AB?

Answer: 5. $AB=\sqrt{(5-1)^2+(5-2)^2}=\sqrt{16+9}=5$.

11 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a rectangle with length 10 and width 3?

Answer: 30. Applying Area, Volume, Triangles, Circles & Trigonometry principles, Area $=l\times w=10\times 3=30$ Therefore, the verified correct option is 30.

12 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Triangle ABC is similar to triangle DEF. Side $AB=10$ corresponds to side $DE=40$, and side $AC=5$ corresponds to side DF. What is the length of DF?

Answer: 20. Scale factor $=\frac{DE}{AB}=\frac{40}{10}$. $DF=AC\times \text{scale}=5\times\frac{40}{10}=20$

13 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Triangle ABC is similar to triangle DEF. Side $AB=9$ corresponds to side $DE=45$, and side $AC=5$ corresponds to side DF. What is the length of DF?

Answer: 25. Scale factor $=\frac{DE}{AB}=\frac{45}{9}$. $DF=AC\times \text{scale}=5\times\frac{45}{9}=25$

14 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A pole casts a shadow 15 feet long. The distance from the top of the pole to the tip of the shadow is 25 feet. How tall is the pole?

Answer: 20. By the Pythagorean theorem: $h=\sqrt{25^2-15^2}=\sqrt{400}=20$ feet.

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Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, $\sin\theta = \frac{3}{5}$. What is $\cos\theta$?

Answer: $4/5$. Adjacent $= 4$. $\cos\theta = \frac{4}{5}$

16 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a triangle with base 15 and height 12?

Answer: 90. Area $= \frac{1}{2}bh = \frac{1}{2}(15)(12) = 90$

17 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has legs of length 3 and 4. What is the length of the hypotenuse?

Answer: 5. $c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

18 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A rectangle has perimeter 36 cm and length = twice width. What is the area in cm²?

Answer: 72. $2(2w + w) = 36 \Rightarrow 6w = 36 \Rightarrow w = 6, l = 12$. Area $= 6 \times 12 = 72$ cm^2

19 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A ladder 29 feet long leans against a wall. The base of the ladder is 20 feet from the wall. How high up the wall does the ladder reach?

Answer: 21. By the Pythagorean theorem: $h = \sqrt{29^2 - 20^2} = \sqrt{441} = 21$ feet.

20 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a $45^\circ - 45^\circ - 90^\circ$ triangle, the hypotenuse is $10\sqrt{2}$. What is the length of each leg?

Answer: 10. In a 45-45-90 triangle, hypotenuse $= s\sqrt{2} \Rightarrow s\sqrt{2} = 10\sqrt{2} \Rightarrow s = 10$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\cos B$?

a. $\frac{4}{5}$ ✓ **b.** $\frac{3}{5}$

c. $\frac{4}{3}$ **d.** $\frac{5}{3}$

$\cos B$ is the adjacent side divided by the hypotenuse. Using the side lengths in the figure, $\cos B = \frac{4}{5}$.

22 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\sin A$?

a. $\frac{5}{4}$ **b.** $\frac{4}{5}$ ✓

c. $\frac{3}{5}$ **d.** $\frac{3}{4}$

$\sin A$ is the opposite side divided by the hypotenuse. Using the side lengths in the figure, $\sin A = \frac{4}{5}$.

23 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\cos A$?

a. $\frac{12}{13}$ **b.** $\frac{5}{12}$

c. $\frac{13}{12}$ **d.** $\frac{5}{13}$ ✓

$\cos A$ is the adjacent side divided by the hypotenuse. Using the side lengths in the figure, $\cos A = \frac{5}{13}$.

24 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the right triangle shown, what is the value of $\sin A$?

a. $\frac{15}{17}$ ✓ **b.** $\frac{8}{17}$

c. $\frac{8}{15}$ **d.** $\frac{17}{15}$

$\sin A$ is the opposite side divided by the hypotenuse. Using the side lengths in the figure, $\sin A = \frac{15}{17}$.

25 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, $\sin\theta = \frac{5}{13}$. What is $\cos\theta$?

a. $\frac{5}{12}$ **b.** $\frac{12}{13}$ ✓

c. $\frac{13}{12}$ **d.** $\frac{5}{13}$

Adjacent $= \sqrt{169 - 25} = 12$. $\cos\theta = \frac{12}{13}$.

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26 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

Two parallel lines are cut by a transversal. One co-interior angle measures 65°. What is the other?

a. 65°

b. 115° ✓

c. 125°

d. 25°

Co-interior angles are supplementary: $180 - 65 = 115^\circ$.

27 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a triangle, two of the angles measure 21° and 59°. What is the measure of the third angle?

a. **100°** ✓

b. 105°

c. 108°

d. 104°

Triangle angles sum to 180°. $180 - 21 - 59 = 100^\circ$

28 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A right triangle has a hypotenuse of length 26 and one leg of length 10. What is the length of the other leg?

a. 30

b. 31

c. 24 ✓

d. 22

$b = \sqrt{26^2 - 10^2} = \sqrt{676 - 100} = \sqrt{576} = 24$

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a triangle, two of the angles measure 38° and 74°. What is the measure of the third angle?

a. 64°

b. **68°** ✓

c. 66°

d. 59°

Triangle angles sum to 180°. $180 - 38 - 74 = 68^\circ$

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30 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a circle with radius 8? (Express your answer in terms of π.)

a. 128π

b. 81π

c. 64π ✓

d. 16π

Applying Area, Volume, Triangles, Circles & Trigonometry principles, Area $= \pi r^2 = \pi(8)^2 = 64\pi$

Therefore, the verified correct option is 64π.

31 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the circumference of a circle with a diameter of 14 inches?

a. 7π in

b. 14π in ✓

c. 28π in

d. 49π in

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $C = \pi d = 14\pi$ in. Therefore, the verified correct option is 14π in.

32 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the radius of the circle with equation $(x-3)^2 + (y+5)^2 = 49$?

a. 5

b. 7 ✓

c. 14

d. 49

In $(x-h)^2 + (y-k)^2 = r^2$, $r^2 = 49 \Rightarrow r = 7$.

33 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the legs have lengths 9 and 12. What is the area of the triangle?

a. 27

b. 54 ✓

c. 72

d. 108

Area $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 9 \times 12 = 54$.

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34 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A square has an area of $64\ \text{m}^2$. What is its perimeter?

a. 16 m

b. 24 m

c. **32 m** ✓

d. 48 m

Side $s = \sqrt{64} = 8$ m. Perimeter $= 4 \times 8 = 32$ m.

35 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the volume of a cylinder with radius 3 cm and height 10 cm?

a. $30\pi\ \text{cm}^3$

b. $60\pi\ \text{cm}^3$

c. $90\pi\ \text{cm}^3$ ✓

d. $120\pi\ \text{cm}^3$

$V = \pi r^2 h = \pi(3^2)(10) = 90\pi\ \text{cm}^3$.

36 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. **96 cm²** ✓

c. 128 cm²

d. 144 cm²

A cube has 6 congruent square faces: $SA = 6s^2 = 6(4^2) = 6(16) = 96\ \text{cm}^2$.

37 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a parallelogram with base 12 cm and height 5 cm?

a. 30 cm²

b. **60 cm²** ✓

c. 90 cm²

d. 120 cm²

Area = base × height $= 12 \times 5 = 60\ \text{cm}^2$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, the side opposite angle θ has length 20, the side adjacent to θ has length 21, and the hypotenuse has length 29. What is $\tan(\theta)$?

a. $121/105$

b. $20/21$ ✓

c. $13/21$

d. $59/84$

$\tan(\theta) = 20/21$ using SOH-CAH-TOA with sides $20, 21, 29$.

39 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the area of a circle with a radius of 6 cm?

a. 12π cm^2

b. 24π cm^2

c. 36π cm^2 ✓

d. 72π cm^2

$A = \pi r^2 = \pi(6^2) = 36\pi$ cm^2.

40 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A rectangle has a width of 7 cm and a diagonal of 25 cm. What is the length of the rectangle?

a. 20 cm

b. 22 cm

c. **24 cm** ✓

d. 26 cm

By the Pythagorean theorem: $l = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ cm.

41 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the center of the circle given by $(x + 2)^2 + (y - 8)^2 = 25$?

a. $(2, -8)$

b. $(-2, 8)$ ✓

c. $(-2, -8)$

d. $(2, 8)$

The center is $(h, k) = (-2, 8)$.

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42 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

What is the volume of a cube with edge length 5 cm?

a. 25 cm³

b. 75 cm³

c. 100 cm³

d. **125 cm³** ✓

$V = s^3 = 5^3 = 125$ cm³.

43 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle, acute angle A measures 30°. What is $\sin(30^\circ)$?

a. $\frac{1}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

From standard 30-60-90 right triangle ratios, $\sin(30^\circ) = \frac{1}{2}$.

44 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle with legs 5 and 12, what is $\tan(\theta)$ if θ is opposite the leg of length 5?

a. $\frac{5}{12}$ ✓

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$.

45 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

The diameter of a circle is 18. What is the area of the circle?

a. 36π

b. 81π ✓

c. 144π

d. 324π

Radius $r = 18/2 = 9$. Area $= \pi r^2 = \pi(9^2) = 81\pi$.

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